Abstract maps and their coordinate matrices

A linear transformation is a map between vector spaces

$$T: V \to W,$$

that preserves addition and scalar multiplication. The map itself does not require a basis. Choosing domain and codomain bases gives its matrix representation.

Rotation matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

is the representation of rotation in standard coordinates. Changing bases changes the entries without changing the underlying rotation.

Kernel, image, rank, injectivity, and surjectivity are intrinsic to the map. Matrix entries depend on coordinates. For determinants, invariance holds under a common basis change for an endomorphism; independent domain and codomain changes alter the coordinate volume factors.

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Linearity

A map $T\colon V \to W$ is linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all vectors and scalars. Equivalently,

$$T(u+v) = T(u) + T(v), \qquad T(cu) = cT(u).$$

Taking scalar $c = 0$ gives

$$T(0) = 0.$$

An affine map that moves the origin,

$$T(x) = Ax + b, \qquad b \neq 0$$

is therefore not a vector-space linear map. Graphics can represent it linearly after adding a homogeneous coordinate.

Linearity preserves every finite linear combination.

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Linearity (continued)

$$T\left(\sum_i c_i v_i\right) = \sum_i c_i T(v_i).$$

Consequently, the images of a basis determine the whole map.

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Kernel, image, and rank-nullity

Define kernel and image by

$$N(T) = \{v \in V : T(v) = 0\},$$
$$C(T) = \{T(v) : v \in V\}.$$

They are subspaces of V and W. For finite-dimensional V, rank-nullity states

$$\dim V = \dim N(T) + \dim C(T).$$

To prove it, take a kernel basis

$$v_1, \ldots, v_k,$$

and extend it to a basis of V.

$$v_1, \ldots, v_k, u_1, \ldots, u_r.$$

The vectors $T(u_1), \ldots, T(u_r)$ span the image by applying T to the full basis expansion. If $\sum_i c_i T(u_i) = 0$, then $\sum_i c_i u_i$ lies in the kernel. Independence of the extended basis forces every $c_i = 0$. Thus these image vectors form a basis.

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Injectivity, surjectivity, and dimension

A linear map is injective exactly when

$$N(T) = \{0\}.$$

If the kernel is trivial, $T(u) = T(v)$ implies $T(u - v) = 0$, hence $u = v$. Conversely, injectivity and $T(v) = T(0)$ imply $v = 0$.

Surjectivity means

$$C(T) = W.$$

For a finite-dimensional map $T : V \to V$, the following are equivalent.

- T invertible;
- The kernel contains only zero.
- The rank equals $\dim V$.
- coordinate matrix invertible.

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Injectivity, surjectivity, and dimension (continued)

With unequal dimensions, injectivity and surjectivity differ. A map $\mathbb{R}^2 \to \mathbb{R}^3$ can be injective, but its image has dimension at most 2 and cannot fill the codomain.

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Examples of linear maps

Rotation:

$$T(x, y) = (-y, x)$$

has zero kernel and the whole plane as its image.

Projection:

$$P(x, y) = (x, 0)$$

has the vertical axis as kernel and the horizontal axis as image.

Polynomial differentiation:

$$D(p) = p'$$

On $\mathcal{P}_k$, with $k \geq 1$, differentiation has the constant polynomials as kernel and $\mathcal{P}_{k-1}$ as image. For $k = 0$, its image is the zero space.

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Examples of linear maps (continued)

Integration:

$$I(p)(t) = \int_0^t p(s)\,ds$$

On $\mathcal{P}_k$, this integration map has trivial kernel. Its image consists of the polynomials of degree at most $k+1$ whose constant term is zero.

Scaling by c is invertible if $c \neq 0$. For $c = 0$, the entire space maps to zero.

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Linear and affine maps

Affine map:

$$T(x) = Ax + b.$$

An affine map preserves affine combinations, $T((1-t)x + ty) = (1-t)T(x) + tT(y)$. A line can collapse to a point if its direction lies in the kernel of A; otherwise its along-line ratios are preserved. For $b \neq 0$,

$$T(0) = b \neq 0,$$

so the map is not linear on the original vector space.

A straight-line graph need not describe a linear map. For example,

$$f(x) = 2x + 1$$

fails $f(0) = 0$ and the scalar multiplication rule.

The map $T(x) = x^2$ is also nonlinear, failing addition and scalar multiplication. Testing $T(0) = 0$ can reject a map immediately, but passing that test alone does not establish linearity.

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Matrix columns from basis images

Input basis:

$$B_{\text{in}} = \{v_1, \dots, v_n\},$$

output basis:

$$B_{\text{out}} = \{w_1, \dots, w_m\}.$$

Expand each $T(v_j)$ in the codomain basis.

$$T(v_j) = \sum_{i=1}^{m} M_{ij} w_i.$$

These coordinate vectors form the columns

$$M = \begin{bmatrix} [T(v_1)]_{B_{\text{out}}} & \cdots & [T(v_n)]_{B_{\text{out}}} \end{bmatrix}.$$

If

$$x = \sum_j c_j v_j,$$

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Matrix columns from basis images (continued)

then

$$[T(x)]_{B_{\text{out}}} = M[x]_{B_{\text{in}}}.$$

The column count equals the domain dimension and the row count equals the codomain dimension. Coordinate order must remain consistent.

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A complete standard-matrix example

Let

$$T(x, y) = (x + y, x - y, 2x).$$

The images of the standard basis vectors are

$$T(e_1) = (1, 1, 2)^T, \qquad T(e_2) = (1, -1, 0)^T.$$

so

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix}.$$

This matrix is 3×2. Its kernel satisfies

$$x + y = 0, \quad x - y = 0, \quad 2x = 0$$

forcing $x = y = 0$, so T is injective. Its image is a rank-2 plane in $\mathbb{R}^3$, not the entire codomain.

Now let $S \colon \mathbb{R}^3 \to \mathbb{R}^2$,

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A complete standard-matrix example (continued)

$$S(x,y,z) = (x+z, y-2z), \qquad N = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}.$$

Composition means $(S \circ T)(v) = S(T(v))$. When the output basis of T matches the input basis of S,

$$[S \circ T]_{\text{std}} = NA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -3 & -1 \end{bmatrix}.$$

For $v = (1,2)^T$, $T(v) = (3,-1,2)^T$, so

$$S(T(v)) = (5,-5)^T = (NA)v.$$

The rightmost factor A acts first, followed by N. Insert a coordinate change if the intermediate bases differ.

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The change-of-basis formula

Let A be the standard-coordinate matrix, with new input and output basis matrices $B_{\text{in}}, B_{\text{out}}$.

$$[x]_{\text{std}} = B_{\text{in}}[x]_{B_{\text{in}}},$$
$$[y]_{B_{\text{out}}} = B_{\text{out}}^{-1} y_{\text{std}}.$$

Apply these operations in order.

1. Convert input basis coordinates to standard coordinates with B_{in}.
2. Apply A.
3. Convert the output to its new basis coordinates with B_{out}^{-1}.

Consequently,

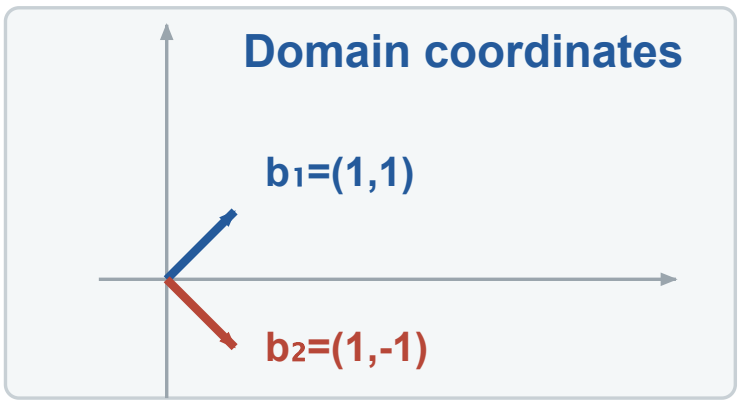
$$[T(x)]_{B_{\text{out}}} = B_{\text{out}}^{-1} A B_{\text{in}} [x]_{B_{\text{in}}},$$
$$M = B_{\text{out}}^{-1} A B_{\text{in}}.$$

The domain and codomain bases can have different dimensions.

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The change-of-basis formula (continued)

Basis conversion: Bin at input, Bout⁻¹ at output



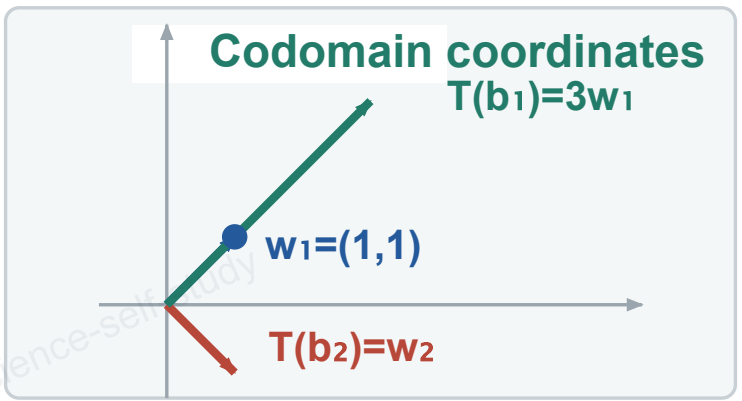
A=[[2,1],[1,2]]

Bin=[[1,1],[1,-1]]

Bout=Bin

M=[[3,0],[0,1]]

M=Bout⁻¹ A Bin



For a two-dimensional example, take

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B_{\text{in}} = B_{\text{out}} = B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Since $Ab_1 = (3,3)^T = 3b_1$、$Ab_2 = (1,-1)^T = b_2$，

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The change-of-basis formula (continued)

$$M = B^{-1}AB = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

for $x = (2, 0)^T$, the input coordinates are $B^{-1}x = (1, 1)^T$. Applying M gives $(3, 1)^T$, and converting back with B gives

$$B(3, 1)^T = (4, 2)^T = A(2, 0)^T.$$

The input conversion uses B_{in}, while the output conversion uses B_{out}^{-1}. These directions must not be reversed.

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Similarity on one space

When $V = W$ and the same new basis B is used on both sides,

$$[T]_B = B^{-1}AB.$$

If $Av = \lambda v$ and $y = B^{-1}v$,

$$B^{-1}AB\,y = B^{-1}Av = \lambda y.$$

so the eigenvalues agree. More fully, $\lambda I - B^{-1}AB = B^{-1}(\lambda I - A)B$, proving equality of characteristic polynomials. Trace and determinant are also invariant.

$$\operatorname{tr}(B^{-1}AB) = \operatorname{tr} A,$$
$$\det(B^{-1}AB) = \det A.$$

Independent input and output bases need not give similarity. The resulting matrix can be rectangular, and even square matrices related by independent left and right changes need not be similar.

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An eigenbasis example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Eigenbasis:

$$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$B^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Therefore

$$B^{-1}AB = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

In this basis, the same map applies independent scalings. An ill-conditioned eigenbasis can amplify coordinate errors, so exact diagonalization does not itself guarantee a good numerical representation.

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Differentiation in two bases

$D\colon \mathcal{P}_2 \to \mathcal{P}_1$, input monomial basis

$$(1, t, t^2),$$

output basis $(1, t)$.

$$D(1) = 0, \qquad D(t) = 1, \qquad D(t^2) = 2t.$$

Hence

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

If the input basis becomes $1, s, s^2$, where $s = t - 1$, then

$$D(s^2) = 2s = -2 + 2t,$$

the new coordinate matrix is

$$\begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

The representation changes, while differentiation itself does not.

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Invariant subspaces

A subspace $U \subseteq V$ is invariant if

$$T(U) \subseteq U.$$

An eigenvector line is a one-dimensional nonzero invariant subspace. If

$$V = U \oplus W$$

and both summands are invariant, the matrix is block diagonal in a basis adapted to the decomposition.

If only U is known invariant, begin a basis with a basis of U. The matrix has block upper triangular form

$$\begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix}.$$

Invariant subspaces reduce a large map to smaller blocks. Both Schur and Jordan forms use this structure.

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Eigenbases and diagonalization

If $b_1, \ldots, b_n$ form an eigenbasis, let

$$B = [b_1 \ \cdots \ b_n],$$

Then

$$B^{-1}AB = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

The eigenbasis separates coupled equations into scalar modes, simplifying powers, exponentials, and recurrences.

For a real symmetric matrix, choose an orthonormal eigenbasis Q, giving

$$Q^{-1} = Q^T.$$

Inversion becomes transposition and the basis condition number is 1. A nonsymmetric eigenbasis can be nearly dependent despite existing exactly.

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Jordan form and generalized eigenvectors

Every complex square matrix has a Jordan form

$$J = B^{-1}AB.$$

whose blocks have the form

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ 0 & \lambda & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 0 & \lambda \end{bmatrix}.$$

The diagonal contains eigenvalues; superdiagonal ones encode generalized-eigenvector chains. A nonzero vector w satisfying

$$(A - \lambda I)^m w = 0$$

for some $m \geq 1$ is a generalized eigenvector. A Jordan chain satisfies $(A - \lambda I)w_1 = 0$, $(A - \lambda I)w_{j+1} = w_j$, giving the superdiagonal ones. The general Jordan existence theorem is stated without its full proof here; the exercises compute a second-order nilpotent block explicitly.

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Jordan form and generalized eigenvectors (continued)

Jordan form is a theoretical classification. Numerical algorithms usually use Schur form, SVD, or structured methods because computing Jordan structure is unstable.

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Circulant matrices and a Fourier basis

For $n = 4$, the left cyclic shift

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

maps $(x_0, x_1, x_2, x_3)^T$ to $(x_1, x_2, x_3, x_0)^T$.

Choose roots of unity

$$\lambda \in \{1, i, -1, -i\},$$
$$v_\lambda = (1, \lambda, \lambda^2, \lambda^3)^T.$$

The wraparound coordinate uses $\lambda^4 = 1$, so

$$Pv_\lambda = \lambda v_\lambda.$$

The four distinct eigenvalues give four independent eigenvectors. The Fourier matrix diagonalizes P and every circulant polynomial

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Circulant matrices and a Fourier basis (continued)

$$C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3.$$

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Useful bases for function spaces

Common finite bases and infinite expansion systems include the following.

- monomials $1, x, x^2, \ldots$:symbolic polynomial operations;
- trigonometric functions:periodic signals;
- Complex Fourier exponentials for translations and convolution.
- Legendre polynomials for orthogonal approximation on $[-1, 1]$.
- Chebyshev polynomials for interpolation and numerical approximation.

The task determines the useful basis. Eigenfunctions can diagonalize a differential operator. A well-conditioned orthogonal basis supports stable approximation. Rapidly decaying coefficients support compression of sufficiently smooth functions. Infinite expansions require a specified convergence sense.

Equal spans mean equal representable spaces. The choice of basis still changes sparsity, conditioning, and interpretation.

Basis conditioning

For

$$X = [x_1 \ \cdots \ x_n]$$

and

$$x = Xc,$$

A coordinate perturbation can be amplified by

$$\|X^{-1}\|_2$$

A large $\kappa_2(X)$ allows a small geometric change to cause a large coordinate change.

This does not make X singular in exact arithmetic. It can be an ill-conditioned basis. An orthonormal basis satisfies

$$\kappa_2(X) = 1.$$

High-degree monomials on large intervals can be poorly conditioned. Scaling, orthogonalization, or Chebyshev bases can help. Record the normalization and condition number of a coordinate change.

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A combined example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$B^{-1}AB = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

For

$$x = (2, 0)^T,$$

eigenbasis coordinates:

$$c = B^{-1}x = (1, 1)^T.$$

$$[T^5 x]_B = \begin{bmatrix} 4^5 & 0 \\ 0 & 2^5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1024 \\ 32 \end{bmatrix}.$$

return to standard coordinates by

$$T^5 x = B \begin{bmatrix} 1024 \\ 32 \end{bmatrix} = \begin{bmatrix} 1056 \\ 992 \end{bmatrix}.$$

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A combined example (continued)

The eigenbasis separates the two modes. An ill-conditioned basis would require an additional coordinate-error analysis.

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Chapter check

You can now do the following.

- Test linearity and use the necessary zero-image condition.
- Prove kernel and image are subspaces and apply rank-nullity.
- Build matrix columns from basis images.
- Use the two-basis matrix formula with the correct multiplication order.
- Distinguish similarity from independent domain and codomain changes.
- Use invariant subspaces, eigenbases, Jordan chains, and Fourier vectors.
- Distinguish exact basis existence from numerical conditioning.

Chapter 9 develops complex inner products, unitary matrices, and FFT.

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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